Skull Fracture Classification - Research Project Report

Introduction

A skull fracture is a head injury which is a break in the bone structure that surrounds the brain. Many skull fractures are the result of an impact or blow to the head. As with any fractures of the bone it occurs when biomechanical stresses exceed the bone's tolerance. It is common to occur after instances of relatively high force such as vehicle accidents, falls, or sports injuries. Skull fractures are more common in children because of their skulls being thinner and more susceptible than those of adults. Adults who are more than the age of 75 are also at a higher risk of skull fractures because of more frequent falls. Skull fractures can result in several critical complications that include intracranial and orbital injuries, cerebrospinal fluid (CSF) leakage, vascular injuries, paralysis, seizures, and blood clots. It can also increase the risk of an underlying subdural and epidural hematoma. Some skull fractures injure arteries and veins which then bleed into the spaces around the brain tissue. Blood may accumulate between the brain and the skull causing an intracranial hematoma. As a result, timely and accurate diagnosis of skull fractures is a necessity for the management of traumatic head injury and mitigating the risk of severe outcomes.
In today's time, in clinical practice, fractures are assessed using medical imaging techniques. Computed tomography (CT) is the commonly used diagnostic tool to treat suspected skull fractures. Radiologists evaluate CT scans of the brain in order to check if a skull fracture is present or not. Radiographs were used for this purpose previously but they have been usurped by CT scans. CT scans are not only sensitive to detect skull fractures but are also able to identify and understand their extent. As a result it enables the possibility for surgical planning. CT is an advanced medical imaging technique that organizes multiple medical X-ray images of a patient from several angles to create slices. It creates 3D images that provide a more comprehensive view than radiographs. In clinical practice CT scans are evaluated by radiologists on a slice-by-slice basis. CT scan evaluation on a slice-by-slice basis manually is rigorous, time consuming, and susceptible to human error. Emergency Departments and radiologists manage a high volume of head injury cases. Skull fractures on CT scans usually appear as discontinuities in the bone and narrow slits. Many skull fractures are small in size, subtle, and microscopic. Some skull fractures may also be overlooked during manual evaluation when they are within the plane of image reconstruction. These factors make the manual diagnosis of skull fractures time consuming and susceptible to human error. As a result it is useful to propose an efficient deep learning classification system for automatic diagnosis of skull fractures. An automatic diagnosis deep learning system will help alleviate radiologists' workload, lower the rate of misdiagnosis, increase the speed of diagnosis, and assist in identifying CT scans in which a skull fracture is present.

Dataset

Bone window CT scan images of 49 patients were collected from a private radiologist in Pakistan. Each patient consisted of CT scan images that contained a fracture or fractures and CT scan images that did not contain a fracture. 44 of the 49 patients contained CT scan images in 5.0mm and 1.3mm slice thickness. 1 patient contained CT scan images of 2.5mm and 0.6mm slice thickness. 1 patient contained CT scan images of 5.0mm and 2.5mm slice thickness. 2 patients contained CT scan images only of 5.0mm slice thickness. The

remaining 1 patient contained CT scan images of only 3.0mm slice thickness. All the collected CT scan images were annotated by the private radiologist. The radiologist annotated all of the CT scan images into CT scan images that contain a fracture and CT scan images that do not contain a fracture. The specific information of the region of fracture/(s) in the CT scan images that contained a fracture/(s) was gathered by the private radiologist for model interpretability. Along with the information the total types of fractures that were present in the CT scan images was also confirmed. The presence and appearance of sutures that resemble a fracture, present in CT scan images that did not contain a fracture, was also looked into and recorded. The dataset comprised a total of 10 types of fractures. The total types of fractures present in the dataset were parietal fractures, frontal fractures, depressed fractures, occipital fractures, temporal fractures, orbital wall fractures, orbital fractures, nasal fractures, blowout fractures, and zygomatic fractures. 7 of the 10 types of fractures included variations of the type itself. These types were parietal, frontal, depressed, occipital, nasal, and blowout, and temporal.

The total number of CT scan images in the dataset was 8,885. The dataset contained 2,051 CT scan images in which a fracture or more than one fracture was present. The remaining 6,834 CT scan images were images in which a fracture was not present. The dataset contained more CT scan images that did not contain a fracture as a result the dataset was imbalanced. The training data consisted of 1,542 “fracture” CT scan images and 6,834 CT scan images that did not contain a fracture. The validation data consisted of 255 “fracture” CT scan images and 1104 CT scan images that did not contain a fracture. The test data consisted of 254 “fracture” CT scan images and 1104 CT scan images that did not contain a fracture.

The data was distributed in training , validation , and test considering each patient’s specific slice version of CT scan images should only be present in either training or validation and test data. This means that for Patient “A” with CT scan images of slice thickness 5.0mm, this specific slice thickness version of CT scan images of Patient “A” will only be present in either the training data or validation and test data. A specific slice thickness version of CT scan images of a patient can, however, be present in the validation or test data if the same patient’s other slice thickness version of CT scan images is present in the training data. The step for implementing this type of distribution was taken to ensure the model being created projects strong generalization capabilities for each type of fracture by identifying unique forms and variations of each fracture and a limited amount of data. The majority of CT scan images in the validation and test data are however of patients that are not present in the training data. A patient’s CT scan images of a scan of a specific thickness and resolution present in the test or validation data for which that patient’s other scan of another thickness and resolution is present in the training data is only for those cases that contain a type of fracture that is limited in number in the data set.

Experimental Environments

Kaggle’s NVIDIA Tesla P100 GPU was used for model training and the model training was executed on Kaggle’s notebook environment. The GPU comprised 4 CPU cores and 29 Gigabytes of RAM. Python was utilized as the programming language and Tensorflow version 2.19.0 was utilised for model building, training, and evaluation.

Methodology

The original format of the CT scan images was DICOM. Due to a lack of computing power the images were converted to JPG format. The image size was reduced to 224x224 and converted to grayscale. The individual pixels in each image were rescaled by dividing with the maximum pixel value.

The task was considered as a binary image classification problem to determine whether or not the CT scan image contains a fracture or does not contain a fracture. Data augmentation was applied only to the fracture CT scan images in the training data to balance both classes of images. Two unique combinations of rotation, horizontal translation, and vertical translation were generated for each fracture CT scan image present in the training data.

The objective of this phase was to design a novel architecture that successfully extracts relevant characteristics from “fracture” CT scan images and “healthy” CT scan images in order to perform correct classifications. As a feature extractor the feature extraction block of the original VGG-16 was used. The feature extraction block of the VGG-16 was identified as a baseline for the novel architecture due to its acceptable performance in accurately classifying CT scans.

Architecture, layers, and parameters of VGG-16

Layer	Activation Shape	Parameters
Input Layer	(224,224,1)	0
CONV2D_1(filter_shape = 3, strides = 1, num_filters = 64, activation = ‘relu’)	(224,224,64)	640
CONV2D_2(filter_shape = 3, strides = 1, num_filters = 64, activation = ‘relu’)	(224,224,64)	36928
MaxPooling2D(filter_shape = 2, strides = 2)	(112,112,64)	0
CONV2D_3(filter_shape = 3, strides = 1, num_filters = 128, activation = ‘relu)	(112,112,128)	73856
CONV2D_3(filter_shape = 3, strides = 1, num_filters = 128, activation = ‘relu)	(112,112,128)	147584
MaxPooling2D(filter_shape = 2, strides = 2)	(56,56,128)	0
CONV2D_5(filter_shape = 3, strides = 1, num_filters = 256, activation = ‘relu’)	(56,56,256)	295168
CONV2D_5(filter_shape = 3, strides = 1, num_filters = 256, activation = ‘relu)	(56,56,256)	590080

CONV2D_5(filter_shape = 3, strides = 1, num_filters = 256, activation = ‘relu’)	(56,56,256)	590080
MaxPooling2D(filter_shape = 2, strides = 2)	(28,28,256)	0
CONV2D_8(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(28,28,512)	1180160
CONV2D_9(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(28,28,512)	2359808
CONV2D_10(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(28,28,512)	2359808
MaxPooling2D(filter_shape = 2, strides = 2)	(14,14,512)	0
CONV2D_11(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(14,14,512)	2359808
CONV2D_12(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(14,14,512)	2359808
CONV2D_13(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(14,14,512)	2359808
MaxPooling2D(filter_shape = 2, strides = 2)	(7,7,512)	0
Flatten	(25088)	0
Dense_1(units = 4096, activation = ‘relu’)	(4096)	102764544
Dropout_1(rate = 0.4)	(4096)	0
Dense_2(units = 4096, activation = ‘relu’)	(4096)	16781312
Dropout_2(rate = 0.4)	(4096)	0
Dense_3(units = 1, activation = ‘sigmoid’)	(1)	4097

Total parameters: 134,263,489

Due to the dataset being highly imbalanced in favour of CT scan images that do not contain a fracture, evaluation metrics such as accuracy were not used for evaluation. Recall and Specificity which evaluate the total number of correct classifications per class were incorporated.

Recall = TP / TP + FN

Specificity = TN / TN + FP

TP = True Positive, FN = False Negative, TN = True Negative, FP = False Negative

Before the start of model training a callback was incorporated based on validation accuracy. The callback was set in accordance to if and only if the validation accuracy is equal to or more than 75% then the weights which generate the highest validation recall must be saved. The callback was incorporated for the purpose of preserving those weights which exhibit , for both , specificity and recall values that project potential to increase. It was important to incorporate the callback and record the weights with the highest recall because the majority of the original images in the dataset were CT scan images that did not contain a fracture. The value of 75% was set because in order for the model to obtain 75% validation accuracy the specificity at a minimum can only be 69%. In the circumstance the model generates 100% validation recall and the validation accuracy is 75% the model would have generated a specificity of 69%. The hyper-parameters that were used can be seen in the table below. Binary cross-entropy was used as the loss function.

Batch size	32
Optimizer	Nesterov Accelerated Gradient with Momentum set to 0.9
Learning Rate Scheduling	Cosine Decay with Warm-up
Initial Learning Rate	1e-2
Final Learning Rate	1e-4
Epochs	300
Warm-up Epochs	5
Convolution Layer Regularization	L2 Regularization 1e-4
Dense Layer Regularization	L2 Regularization 1e-3
Dropout	0.4

Although the VGG-16 network showed acceptable results it produced incorrect classifications for many fracture CT scan images. After observing the Grad-CAM++ heatmaps for the False Negatives it showed that the region of pixels which constitute a fracture were considered but an incorrect classification was made. One of the reasons for this was due to VGG-16 considering those regions as well that are present in healthy CT scan images such as background, lighting, skull shape, orientation, outer table, inner table, patient credentials, and ct scan information. Another reason for this was that many healthy CT scan images contain sutures that have a similar appearance to fractures. A factor that also played a key role in the above results was that the overall dataset was highly imbalanced in favor of healthy CT scans. In order to balance the training data, data augmentation was implemented only for the fracture class in order to balance the training data.

Considering the above results and observations a novel architecture is proposed which does not contain a flatten layer after extracting the features from the convolutional layers but aims to reduce the redundant features while preserving the relevant regions to fracture CT scan images.

Architecture, layers, and parameters of VGG-16 with Gated Multiplication

Layer	**Activation Shape**	**Parameters**
Input Layer	(224,224,1)	0
CONV2D_1(filter_shape = 3, strides = 1, num_filters = 64, activation = ‘relu’)	(224,224,64)	640
CONV2D_2(filter_shape = 3, strides = 1, num_filters = 64, activation = ‘relu’)	(224,224,64)	36928
MaxPooling2D(filter_shape = 2, strides = 2)	(112,112,64)	0
CONV2D_3(filter_shape = 3, strides = 1, num_filters = 128, activation = ‘relu)	(112,112,128)	73856
CONV2D_3(filter_shape = 3, strides = 1, num_filters = 128, activation = ‘relu)	(112,112,128)	147584
MaxPooling2D(filter_shape = 2, strides = 2)	(56,56,128)	0
CONV2D_5(filter_shape = 3, strides = 1, num_filters = 256, activation = ‘relu’)	(56,56,256)	295168
CONV2D_5(filter_shape =	(56,56,256)	590080

3, strides = 1, num_filters = 256, activation = ‘relu)		
CONV2D_5(filter_shape = 3, strides = 1, num_filters = 256, activation = ‘relu’)	(56,56,256)	590080
MaxPooling2D(filter_shape = 2, strides = 2)	(28,28,256)	0
CONV2D_8(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(28,28,512)	1180160
CONV2D_9(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(28,28,512)	2359808
CONV2D_10(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(28,28,512)	2359808
SpatialDropout2D_1(rate = 0.1)	(28,28,512)	0
MaxPooling2D(filter_shape = 2, strides = 2)	(14,14,512)	0
CONV2D_11(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(14,14,512)	2359808
CONV2D_12(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(14,14,512)	2359808
CONV2D_13(filter_shape = 3, strides = 1, num_filters = 512, activation = ‘relu’)	(14,14,512)	2359808
SpatialDropout2D_1(rate = 0.1)	(14,14,512)	0
MaxPooling2D(filter_shape = 2, strides = 2)	(7,7,512)	0
GlobalMaxPooling2D()	(512)	0
GlobalAveragePooling2D()	(512)	0
Dense_1_GlobalMaxPooling_Projection(units = 512, activation = ‘relu’)	(512)	262656
Dense_2_GlobalAveragePo	(512)	262656

oling_Projection(units = 512, activation = ‘sigmoid’)		
GatedMultiplication_Global MaxPoolingProjection_and_ GlobalAveragePoolingProje ction_Layer	(512)	0
Dropout_1(rate = 0.5)	(512)	0
Dense_3(units = 1, activation = ‘sigmoid’)	(1)	513

Total parameters: 15,239,361

Before the start of model training a callback was incorporated based on validation accuracy. The callback was set in accordance to if and only if the validation accuracy is equal to or more than 80% then the weights which generate the highest validation recall must be saved. The callback was incorporated for the purpose of preserving those weights which not only exhibit an acceptable specificity but also an acceptable recall. It was important to incorporate the callback and record the weights with the highest recall because the majority of the original images in the dataset were CT scan images that did not contain a fracture.
The value of 80% was set because in order for the model to obtain 80% validation accuracy the specificity at a minimum can only be 75%. In the circumstance the model generates 100% validation recall and the validation accuracy is 80% the model would have generated a specificity of 75%.

The replacement of the Flattened layer and the 4096 neuron dual dense layers with the Gated Multiplication Mechanism constituting the features extracted by the Global Max Pooling and Global Average Pooling layers enabled the possibility of enhancing recall and eliminated redundant features that were being extracted from the convolutional layers.

Global Max Pooling extracts the single highest pixel value from each of the 7x7x512 feature maps that were generated by the fifth convolutional block. For a single image the operation generates 512 features as compared to the original VGG-16 which generated 25,088 features. As a result, redundant features were eliminated and the single highest pixel information was recorded. For fracture CT scan images the information of the fracture region was preserved. For a healthy CT scan image the single highest pixel value for each of the 7x7x512 feature maps was preserved as well.

<u>Global Max Pooling of a 7x7x512 feature map after fifth convolutional block:</u>

0.1	0.4	1.2	4.3	5.1	6.7	0.1

0.2	0.3	0.5	0.4	0.9	7.0	5.4
0.2	0.3	8.0	10.0	9.0	4.1	4.5
0.1	0.4	5.0	7.8	4.3	5.6	5.0
0.3	3.9	3.0	6.4	6.6	5.2	4.3
0.2	0.4	1.4	1.3	6.5	5.3	4.7
1.2	2.4	2.9	8.7	5.9	6.6	4.2

Global max pooling output = max(Feature Map Pixels)

Global Average Pooling and Gated Multiplication play a key role in identifying a non-fracture region from a fracture. While Global Max Pooling preserves the fracture signal for fracture CT scan images, for healthy CT scan images the single highest pixel value can preserve information of a suture that can appear to be a fracture or a non-fracture part of the image such as an inflammation in the outer table or inner table. However, Global Average Pooling takes the average value of the 7x7 feature map which along with the fracture region also preserves the surrounding context. The surrounding context of many fractures in the dataset is visibly irregular or surrounded with a crack and for many sutures and inflammations is uniform. As a result a distinction between such fractures is created with healthy CT scan images that contain sutures or inflammations. The 512 neurons Global Max Pooling projection dense layer is used to filter irrelevant peaks and ensure correct weights are given to correct features. The 512 neurons Global Average Pooling projection dense layer is used to ensure correct weights are given to correct surroundings for both fracture and healthy CT scan images. The projection dense layer uses the sigmoid activation function which is especially useful for cases where a fracture has a subtle but visible difference with a suture or inflammation as it widens the mathematical difference. Mathematically sigmoid amplifies small differences. As a result the close single highest pixel values when operated through the Global Average Pooling projection layer with sigmoid are widened.

Global Average Pooling of a 7x7x512 feature map after fifth convolutional block:

0.1	0.4	1.2	4.3	5.1	6.7	0.1
0.2	0.3	0.5	0.4	0.9	7.0	5.4
0.2	0.3	8.0	10.0	9.0	4.1	4.5
0.1	0.4	5.0	7.8	4.3	5.6	5.0
0.3	3.9	3.0	6.4	6.6	5.2	4.3

0.2	0.4	1.4	1.3	6.5	5.3	4.7
1.2	2.4	2.9	8.7	5.9	6.6	4.2

Global average pooling output = avg(Feature Map Pixels)

Gated Multiplication classification for a fracture that has a clean surrounding with a subtle difference from a suture that has a clean surrounding:

Fracture Image:

Global Max Pooling Projection Layer Fracture Signal Value: 9.5

Context Average (Global Average Pooling): 0.50

Suture or Inflammation Image:

Global Max Pooling Projection Layer Suture/Inflammation Signal Value: 9.0

Context Average (Global Average Pooling): 0.25

The Global Average Pooling Projection Layer Calculation

Let's assume the learned parameters for the Global Average Pooling Projection Layer are:

Weight**:** 8.0
Bias**:** -3.0

Calculation for Fracture Image

Input from Global Average Pooling Layer: 0.50

$Z = (8.0 \times 0.50) - 3.0$

$Z = 4.0 - 3.0$

$Z = 1.0$

Now, apply Sigmoid:

Sigmoid(1.0) = 0.73

Calculation for Suture/Inflammation Image

Input from Global Average Pooling Layer: 0.25

$Z = (8.0 \times 0.25) - 3.0$

$Z = 2.0 - 3.0 = -1.0$

$Z = -1.0$

Now, apply Sigmoid:

Sigmoid(-1.0) = 0.27

The Gated Multiplication Layer Calculation

Now we multiply the values from the Global Max Pooling Projection Layer by the Global Average Pooling Projection Layer values above.

Fracture Signal from Fracture Image

Signal (fracture) = 9.5×0.73

Signal (fracture) = 6.93

Suture/Inflammation Signal from Suture/Inflammation Image

Signal (suture/inflammation) = 9.0×0.27

Signal (suture/inflammation) = 2.43

The Final Dense Layer Prediction (The Forward Pass)

Let's assume the learned parameters for the Dense Layer are:

Weight**:** 0.5
Bias**:** -3.0

Calculation for Fracture Image

Signal (fracture): 6.93

$Z = (0.5 \times 6.93) - 3.0$

$Z = 3.465 - 3.0$

$Z = 0.465$

Now, apply Sigmoid:

Sigmoid(0.465) = 0.61

The generated probability is 0.61 which is a classification of 1 that points to ‘fracture’

Calculation for Suture/Inflammation Image

Signal (suture/inflammation): 2.43

$Z = (0.5 \times 2.43) - 3.0$

$Z = 1.215 - 3.0$

$Z = -1.785$

Now, apply Sigmoid:

Sigmoid(-1.785) = 0.14

The generated probability is 0.14 which is a classification of 0 that points to ‘no-fracture’

The hyper-parameters that were used can be seen in the table below. Binary cross-entropy was used as the loss function with the modification that when a misclassification of a fracture CT scan image occurs the loss generated from the image will be multiplied by a weight of 3. The class-weights for the fracture class were set to 3 and the weights for the healthy class remained 1. The modification in the loss function was integrated for the purpose of enhancing recall. Spatial Dropout2D was integrated to the architecture as well in the fourth and fifth convolutional blocks for the purpose of removing redundant feature maps.

Batch size	32
Optimizer	Nesterov Accelerated Gradient with Momentum set to 0.9

Learning Rate Scheduling	Cosine Decay with Warm-up
Initial Learning Rate	5e-3
Final Learning Rate	1e-5
Epochs	300
Warm-up Epochs	5
Convolution Layer Regularization	L2 Regularization 1e-3
Dense Layer Regularization	L2 Regularization 1e-3
Dropout	0.5

Experimental Results

The standard VGG-16 model exhibited a validation recall of 76.86% and a test recall of 73.62%. It exhibited a validation specificity of 84.8% and a test specificity of 92.3%. After verifying the True Positives through Grad-CAM++, by observing if in the heatmaps of the True Positives the pixels of the region of the fracture are considered in the classification decision, the final validation recall and test recall values were 76.07% and 71.65% respectively.

Class	**Number of Images in Validation set**	**True Classifications in Validation set**	**Incorrect Classifications in Validation set**
Fracture	255	194	61
No-Fracture	1104	937	167

Class	**Number of Images in Testing set**	**True Classifications in Testing set**	**Incorrect Classifications in Testing set**
Fracture	254	182	72
No-Fracture	1104	1019	85

The proposed VGG-16 Gated Multiplication model exhibited a validation recall of 89.41% and a test recall of 81.88%. It exhibited a validation specificity of 80.16% and a test specificity of 80.70%. After verifying the True Positives through Grad-CAM++, by observing if in the heatmaps of the True Positives the pixels of the region of the fracture are considered

in the classification decision, the final validation recall and test recall values were 87.45% and 80.31% respectively.

Class	**Number of Images in Validation set**	**True Classifications in Validation set**	**Incorrect Classifications in Validation set**
Fracture	255	223	32
No-Fracture	1104	885	219

Class	**Number of Images in Testing set**	**True Classifications in Testing set**	**Incorrect Classifications in Testing set**
Fracture	254	205	50
No-Fracture	1104	891	213

Discussion

The model is able to classify all types of fractures correctly. The limitation of the model exists in fractures with extremely small gaps that are filled with light and contrast which make those fracture CT scan images look similar to “no-fracture” images. The model exhibits a limitation for those “fracture” CT scan images as well which have a fractured region appearance not present in the training data. The limitation for “no-fracture” images exists in the majority for many images that contain a suture still in spite of the fact it is able to distinguish between some images that contain a suture or inflammation looking similar to a fracture in “no-fracture” CT scan images in the dataset. VGG-16 was used as a feature extractor due to its small filters and stride for localizing fractures instead of AlexNet and ResNet which have large filters not able to localize fractures accurately.